Take-Home Assignment: VC Audit Tool

- **Timeline:** 24-48 hours
- **Deliverables:** Working code in a public GitHub repo + short README (max 1 page). Includes instructions for running.
- **Follow-up:** 30-minute walkthrough interview covering design, reasoning, and technical implementation

Problem Overview

Auditors reviewing venture capital portfolios need to estimate the fair value of **private, illiquid portfolio companies**. The goal of this exercise is not to produce a "correct" valuation, but to design and implement a structured, auditable workflow that helps auditors efficiently arrive at a consistent, well-documented fair value estimate.

This task is distinct from valuing publicly traded companies, which have a readily available market price. Private companies lack this price and often have sparse or non-standardized financial data. The challenge is to build a system that can value these private assets using the best available data, which might include internal company metrics, data from the last funding round, or data from publicly-traded comparable companies.

Your solution should demonstrate how you translate a business problem into a sound, maintainable backend system.

Potential Methodologies (for inspiration)

A key part of this exercise is choosing a valuation methodology and building a structured workflow around it. Your system should be able to ingest data, apply a chosen model, and output the result. We are not testing your financial modeling expertise, but rather your ability to translate one of these business-logic workflows into code.

You are free to choose your approach. Examples of methodologies you could model include:

1. **Comparable Company Analysis (Comps):**
    a. **Logic:** The system identifies a set of comparable public companies (based on sector, size, etc.).

b. **Data:** It would fetch their financial data (e.g., Revenue, EBITDA) and Enterprise Value.
c. **Calculation:** It determines a relevant valuation multiple (e.g., Enterprise Value / Revenue).
d. **Application:** It then applies this average multiple to the private portfolio company's own revenue (e.g., 8x EV/Revenue * $10M Revenue = $80M Fair Value).

2. **Discounted Cash Flow (DCF):**
    a. **Logic:** The system projects a company's future cash flows and discounts them back to their present value.
    b. **Data:** This approach would assume you are given a set of detailed financial projections for the portfolio company (e.g., 5-year forecasts for revenue, expenses, and capital expenditures).
    c. **Calculation:** The system would apply a discount rate (e.g., a WACC) to these future cash flows.
    d. **Application:** The sum of the discounted cash flows would be the estimated fair value.
3. **Last Round (Market-Adjusted) Valuation:**
    a. **Logic:** This is a common "mark-to-market" approach. It starts with the valuation from the company's most recent funding round and adjusts it based on public market performance.
    b. **Data:** The inputs would be the company's last_post_money_valuation, the date_of_last_round, and a data source for a relevant public index (e.g., the Nasdaq Composite).
    c. **Calculation:** The system calculates the percentage change in the public index from the date_of_last_round to the present.
    d. **Application:** This percentage change is then applied to the last_post_money_valuation to produce a new, auditable fair value estimate.

Requirements

Inputs

- Portfolio company name (e.g., "Basis AI", "Inflo")
- **Key data required by your chosen methodology.** Examples:
    - For Comps: sector, revenue
    - For DCF: financial_projections (could be a path to a file or a JSON object)
    - For Last Round: last_post_money_valuation, last_round_date

Outputs

- Estimated fair value (numeric or range)
- Key inputs and assumptions used (e.g., "Used 8.2x EV/Revenue multiple", "Used 15% discount rate")
- Citations or data sources (e.g., "Public comp data from Yahoo Finance API [mocked]", "Internal projections dated Q3 2025")
- Explanation of how the estimate was derived (structured or narrative, e.g., "Valuation based on peer-group median multiple of 8.2x applied to $10M LTM Revenue.")

Core Expectations

- Backend service that takes an input and returns the valuation output.
- **Strong code quality, organization, and documentation.**
- Clear workflow from data ingestion → valuation logic → output.
- **Traceable, auditable process:** It should be easy to understand *how* the result was produced.

Preferred

- Structured approach to problem solving (clear steps, modular design).
- If external data is used (e.g., public APIs, datasets), handle errors gracefully (mocking is perfectly fine).
- Thoughtful UX – CLI tool is fine, or simple web frontend. Whatever you need to communicate how auditors would use this in practice. Wireframes/sketches also acceptable.

Not Required

- Extensive deployment setup
- Highly accurate financial modeling (the logic is more important than the numbers)

Deliverables

1. Source code in a public GitHub repository

2. README including:

- Overview of the problem and your **chosen approach/methodology**.

- Key design decisions and tradeoffs.
- Setup instructions and usage example.
- Notes on potential improvements if you had more time.
- (Optional) Screenshot or diagram of UX/workflow.

3. Set up time with us to review live